

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]
Dec-25-0648

MCA-7302 (Analysis and Design of Algorithms)
MCA-3rd NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section-A

1. Write the algorithm to sort an array of integers using quicksort sorting technique. Also, discuss the working of the algorithm with the help of an example. (12)
2. Discuss in detail the concept of efficiency of an algorithm. Explain various asymptotic notations used for expressing the complexity of an algorithm. (12)

Section-B

3. Write the breadth first search algorithm to traverse a graph. Also, discuss the working of the algorithm with the help of an example. (12)
4. Determine the lower bound for comparison-based sorting algorithms using decision tree. (12)

Section-C

5. Write and explain a Monte Carlo randomized algorithm for primality testing. (12)

2

MCA-7302

6. Explain the difference between deterministic and non-deterministic algorithm. Write a non-deterministic sorting algorithm to illustrate the difference. (12)

Section-D

7. Explain the working of Rabin-Karp algorithm for string matching with the help of an example. Also, discuss its limitations. (12)
8. What is finite Automata? Discuss in detail how it can be used for string matching? Explain its advantages. (12)

Section-E (Compulsory)

9. (a) Discuss briefly the working of Branch and Bound algorithm.
(b) Explain the subset paradigm of greedy technique.
(c) What is a minimum spanning tree? Give example.
(d) What are NP-complete and NP-Hard problems? Explain.
(e) What is polynomial time reduction? Explain.
(f) Briefly discuss the applications of string matching algorithms. (6×2=12)